Department of Computer Science and Engineering (CSE)

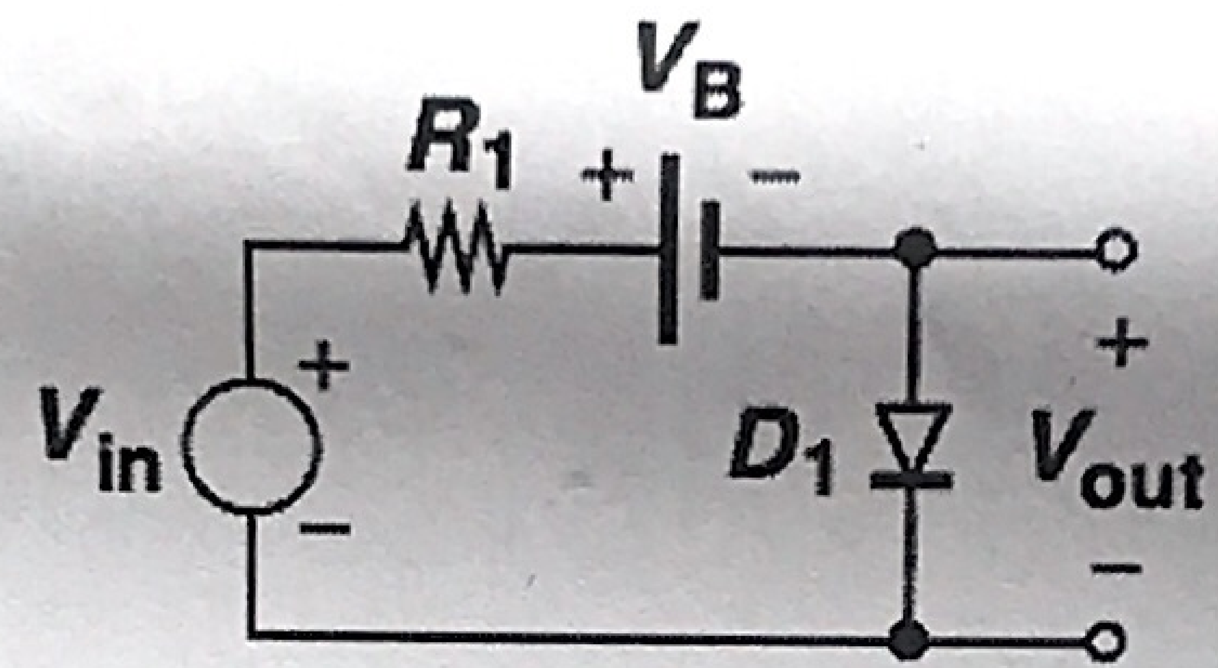
Bangabandhu Sheikh Mujibur Rahman University, Kishoreganj

2nd Year 1st Semester Mid Examination, 2024

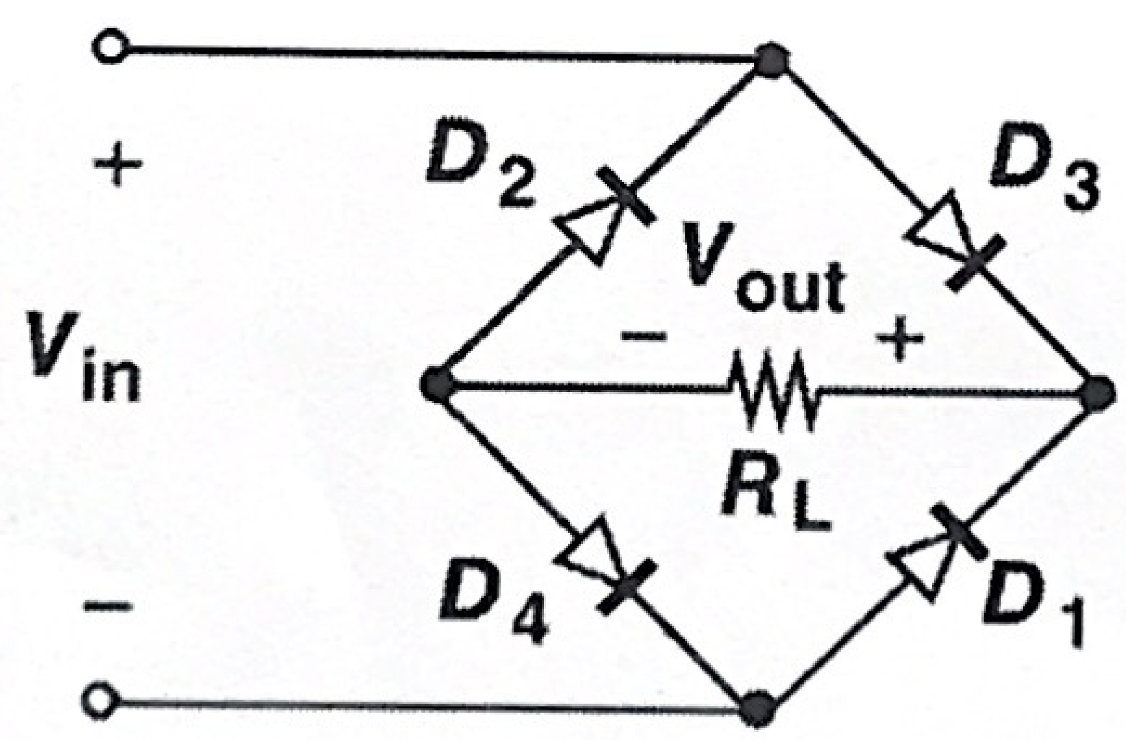
Course Code: EEE-2109 Course Title: Electronics

Total Marks: 25 Duration: 1 hour

1\. (a) Plot the input/output characteristics of the circuit shown in Fig. below using an ideal model for the diodes. 6



(b) Assuming a constant-voltage model for the diodes, plot the input/output characteristic of a full-wave rectifier circuit depicted in Figure 1. 6.5



2\. (a) Draw the circuit diagram of a half-wave rectifier circuit and describe its operation. 6

(b) Given the scenario of a cellphone receiving a signal from a base station, where the signal strength increases as the user approaches the base station, there is a risk that the signal level may become large enough to saturate the circuits within the receiver chain. To prevent saturation and ensure proper functioning, it is crucial to limit the signal amplitude. Design a circuit to limit the signal amplitude to ±100 mV as the user comes closer to the base station, with the diode turn-on voltage, $V_{D.on.}$ set at 800 mV. 6.5